

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech., Winter 2017 - 18 Examination

Semester: 5
Subject Code: 03105303
Subject Name: Theory of Computation

Date: 27/12/2017
Time: 10:30am to 1:00pm
Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions - (All are compulsory) (Each of one mark)**(15)**

1. The transition function of NFA is _____.
2. Define Recursive Enumerable Set.
3. Define Infinite language.
4. What will be the universal language defined over alphabet {a, b}?
5. The number of languages accepted by NPDA and DPDA is same. True / False
6. Which grammar is called as an Ambiguous grammar?
7. Let W is a string of length 3 over the alphabet $\Sigma = \{a\}$ then the number of substrings of W is _____.
8. A regular grammar is
 A) Type-0 B) Type-1 C) Type-2 D) Type-3
9. Every subset of a regular language is regular. True / False
10. There are _____ tuples in PDA.
11. What is the meaning of string acceptance by Empty Stack in PDA?
12. Every finite language is Regular language. True / False
13. Give one example of Left Recursive Grammar.
14. Define ϵ - closure of a state.
15. The language $L = \{a^n b^n \mid n \geq 1\}$ is regular language. True / False

Q.2 Answer the following questions. (Attempt any three)**(15)**

- A) For all positive integers n, prove that $1 + 2 + 3 + \dots + n = n(n + 1)/2$
- B) Define Pumping Lemma and prove the following language is not regular.
 $L = \{a^n b^n \mid n \geq 1\}$

C) Minimize the following DFA

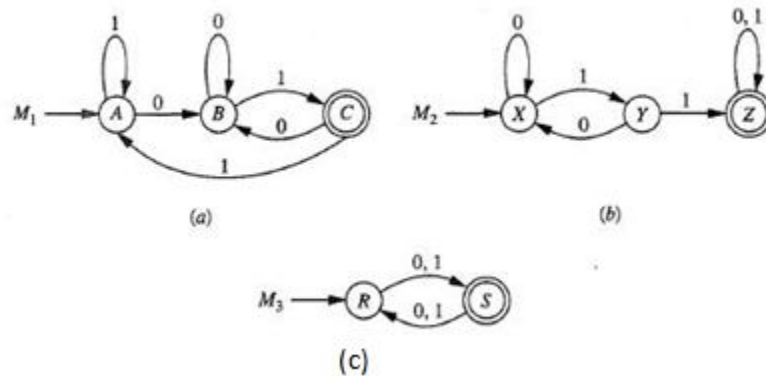
Sate	$\delta(q,0)$	$\delta(q,1)$
A(Initial state)	B	C
B	A	D
C(Final state)	E	F
D(Final state)	E	F
E(Final state)	E	F
F	F	F

D) Convert the following Mealy Machine into Moore Machine

State	$\delta(q,0)$	$\lambda(\text{output})$	$\delta(q,1)$	$\lambda(\text{output})$
A(Initial State)	D	0	C	1
B	C	1	E	1
C	E	0	A	0
D	A	0	B	0
E	B	1	D	0

Q.3 A)

(07)



Let M_1 , M_2 and M_3 be the FAs pictured in Figure above, recognizing languages L_1 , L_2 and L_3 respectively.

Draw FAs recognizing the following languages.

- (1) $L_1 \cup L_2$
- (2) $L_1 \cap L_3$
- (3) $L_2 - L_1$

B) (1) Draw Turing Machine for $L = \{ a^n b^n c^n \mid n \geq 1 \}$ (5 Marks)

(08)

(2) Explain Universal Turing Machine.

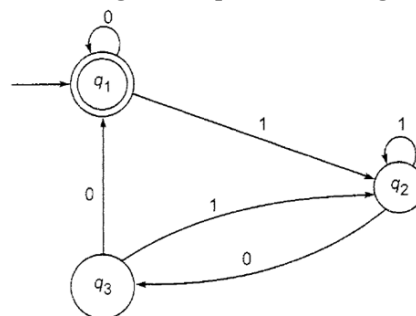
(3 Marks)

OR

B) (1) For the following Regular Expression draw an NFA- Λ recognizing the corresponding languages. $(00 + 1)^* (10)^* 1^*$ (4 Marks)

(08)

(2) Convert the following FA into Regular Expression using Arden's Theorem: (4 Marks)



Q.4 A) Convert the following CFG into CNF and GNF

(07)

$S \rightarrow AaA \mid CA \mid BaB$
 $A \rightarrow aaBa \mid CDA \mid aa \mid DC$
 $B \rightarrow bB \mid bAB \mid bb \mid aS$
 $C \rightarrow Ca \mid bC$
 $D \rightarrow bD \mid \Lambda$

OR

A) Write Formal definition of PDA. Draw PDA for the following language

(07)

$L = \{ WW^R \mid W \in \{a, b\}^* \text{ and } W^R \text{ is reverse of } W \}$

B) (1) Explain the CHOMSKY classification of Grammar.

(4 Marks)

(08)

(2) Give CFG for the following languages

(4 Marks)

- (a) $L_1 = \{ a^n b^n \mid n \geq 1 \}$
- (b) $L_2 = \{ a^i b^j c^k \mid i > j+k \}$